

Classical Mechanics & Special Theory of Relativity

Lecture #8

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Scattering From A Central Potential:

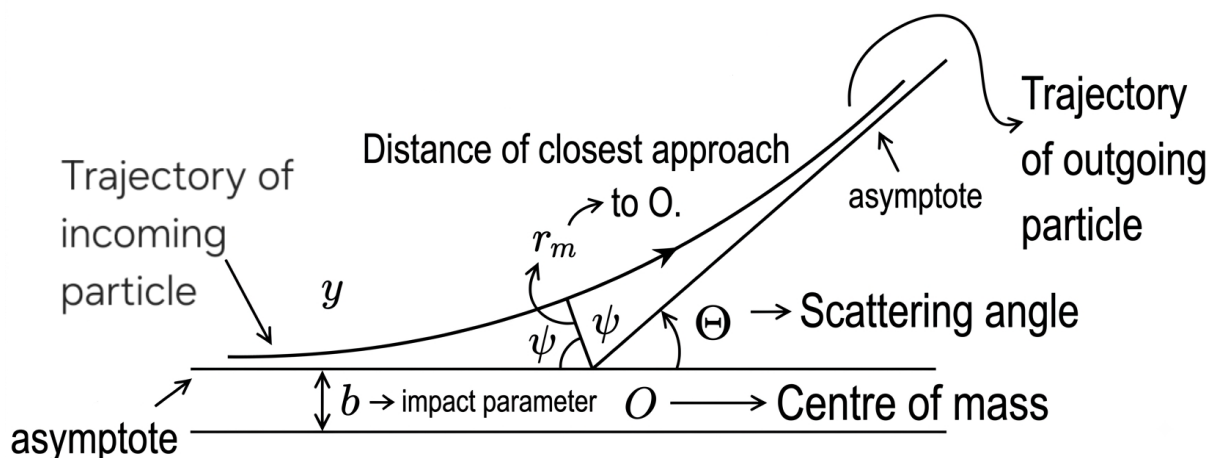
The two body central force problem is not only important in planetary motion but also in atomic physics. The first spectacularly successful model of the hydrogen atom was proposed by Niels Bohr and it was based on the planetary model. Bohr added a few ad-hoc assumptions to it such as the electron orbits are stationary and non-radiating, and that the angular momenta of the orbits are quantized.

The dynamics of unbounded particles are also important in atomic and molecular physics as it allows us to probe the forces that allow atoms and molecules to stay together. In the quantum regime elastic and inelastic collisions are important for exploring the structure of the fundamental fields that make up the universe.

For now, we consider the problem of two classical particles colliding and the determination of the angle of scattering Θ . For this we go to the centre of mass frame with a particle with reduced mass μ moving in a central potential field $V(r)$.

If Ψ is the angle from the centre to the incoming particle then they are related by

$$\Theta = |\pi - 2\Psi|$$



As we have seen the angle formula is given by

$$\begin{aligned}\theta &= \int_{r_0}^r \frac{dr'}{r'^2 \sqrt{\frac{2mE}{l^2} - \frac{2mV(r)}{l^2} - \frac{1}{r'^2}}} + \theta_0 \\ &= \int_{r_0}^r \frac{l dr'}{r'^2 \sqrt{2m(E - V(r)) - l^2/r^2}}\end{aligned}$$

We get for Ψ :

$$\Psi = \int_{r_m}^{\infty} \frac{l dr}{r^2 \sqrt{2m(E - V(r)) - l^2/r^2}}$$

where r_m is the distance of closest approach.

For motion where the incident particles are coming in from infinity it is easier to use the velocity v_{∞} at infinity and the impact parameter b of the particle instead of the energy E and the angular momentum l .

The impact parameter b is defined to be the perpendicular distance from the centre to the direction of v_{∞} . I.e., it is the distance at which the particle would pass the centre if there were no field of force.

In terms of b and E , the new quantities are:

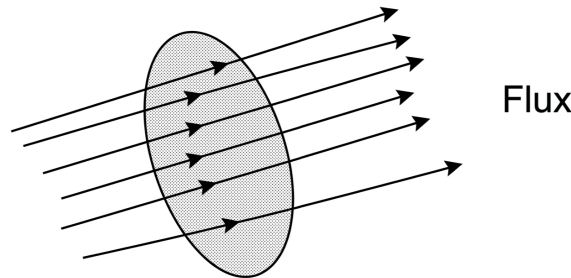
$$l = mv_{\infty}b = b\sqrt{2mE}$$

[This follows from the conservation of angular momentum and energy.]

Note that if b and v_{∞} are fixed (alternatively E and l are fixed) the scattering angle Θ is uniquely determined.

In physical applications we are concerned with a beam of identical particles with uniform velocity v_{∞} being incident on the central potential. The particles only differ in their impact parameters.

We assume that different particles that have different impact parameters are then deflected by different angles. Let us define by I , the intensity or the flux density of the incoming beam. It is the number of particles that cross a unit area perpendicular to the direction of motion per unit time:



If $\vec{\Omega}$ is a particular direction then the differential cross-section for scattering through a small solid angle $d\Omega$ in the direction $\vec{\Omega}$ is defined by

$$\sigma(\vec{\Omega})d\Omega = \frac{\text{number of particles scattered in solid angle } d\Omega \text{ per unit time in the direction } \vec{\Omega}}{\text{Incident Intensity } I}$$

For central forces, there must be symmetry around the axis of the incident beam (why?) and so we can do circle integration in $d\Omega$ and find

$$d\Omega = 2\pi \sin \Theta d\Theta$$

For simplicity we shall assume that diff. impact parameters cannot lead to the same scattering angle. [Not true in general.] Therefore, the number of particles scattered into the solid angle $d\Omega$ between angles Θ & $\Theta + d\Theta$ must be equal to the number of incident particles with impact parameter lying between b and $b + db$:

$$2\pi b I |db| = 2\pi \sigma(\Theta) I \sin \Theta d\Theta$$

Assuming $b = b(E, \Theta)$ one can derive

$$\sigma(\Theta) = \frac{b}{\sin \Theta} \left| \frac{db}{d\Theta} \right|$$

To obtain explicit expression for the differential cross section we assume:

$$F = \frac{zz'q^2}{r^2}$$

$$V(r) = -\frac{k}{r} \quad (k = -zz'q^2)$$

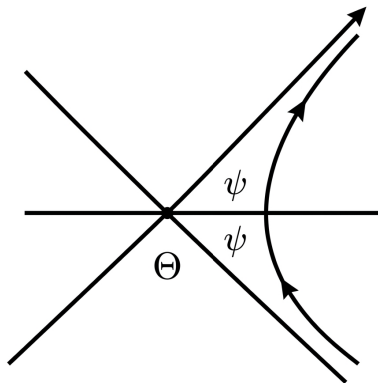
which represents a repulsive Coulomb force between two positively charged particles of charge zq and $z'q$.

Then the solution to the orbit equation gives: Recall the equation of the orbit:

$$\frac{1}{r} = \frac{mk}{l^2} \left(1 + \sqrt{1 + \frac{2El^2}{mk^2}} \cos(\theta - \theta') \right)$$

We choose $\theta' = \pi$ then $\cos(\theta - \pi) = -\cos \theta$ and the equation for orbit becomes

$$\frac{1}{r} = +\frac{mzz'q^2}{l^2} (e \cos \theta - 1)$$



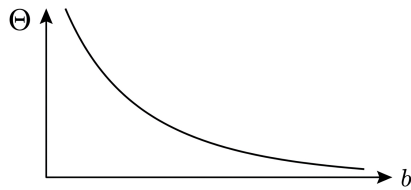
where

$$e = \sqrt{1 + \frac{2El^2}{m(zz'q^2)^2}} = \sqrt{1 + \left(\frac{2E \cdot b}{zz'q^2}\right)^2}$$

The maximum of $\frac{1}{r}$ and hence $r = r_m$ occurs at $\cos \theta = 1 \Rightarrow \theta = 0$. And so $\theta = 0$ corresponds to $r = r_m$. This point is known as the periapsis.

For $r = \infty$ we obtain $\theta = \Psi$ and so

$$\begin{aligned}\cos \Psi &= \frac{1}{e} \\ \Rightarrow \sin \frac{\Theta}{2} &= \frac{1}{e} \\ \cot^2 \frac{\Theta}{2} &= e^2 - 1 \\ \cot \frac{\Theta}{2} &= \frac{2Eb}{zz'q^2} \\ b &= \frac{zz'q^2}{2E} \cot \frac{\Theta}{2}\end{aligned}$$



The differential cross-section is then computed to be

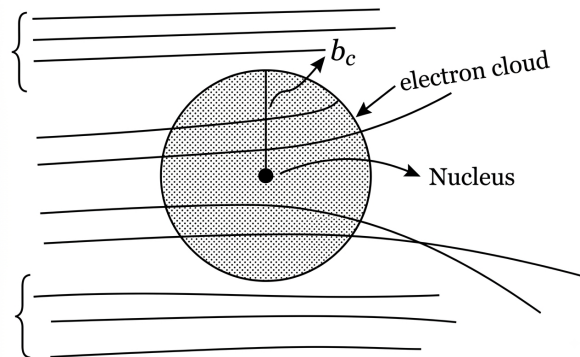
$$\sigma(\Theta) = \frac{1}{4} \left(\frac{zz'q^2}{2E} \right)^2 \operatorname{cosec}^4 \frac{\Theta}{2}$$

which is the Rutherford Scattering cross-section.

The total cross-section is then

$$\sigma_T = 2\pi \int_0^\pi \sigma(\Theta) d\Theta \rightarrow \infty$$

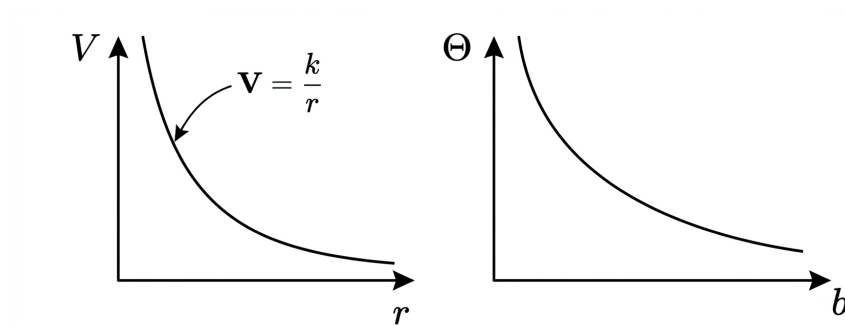
The divergence of the total cross-section originates from the fact that the Coulomb force has an infinite range. Thus particles with values of b very large still get scattered. If for some reason the particles outside a certain cut-off impact parameter b_c were not scattered then σ_T would be finite.



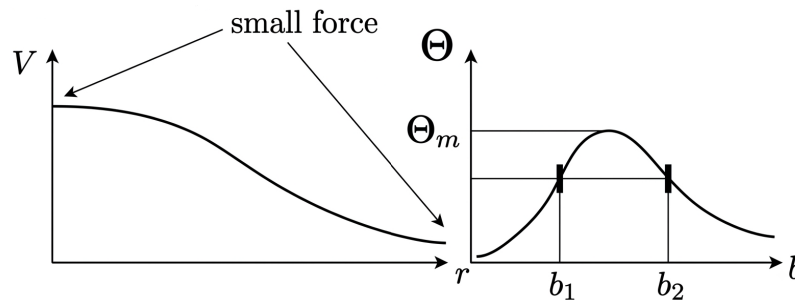
The particles that pass far outside the electron cloud do not "see" the Coulomb force.

Discussion on General Scattering

We assumed that for a given value of b , impact parameter we had one possible value of Θ . This is true of the coulomb potential $V = \frac{k}{r}$.



This is because the force is different for each value of r . But if that weren't the case, In



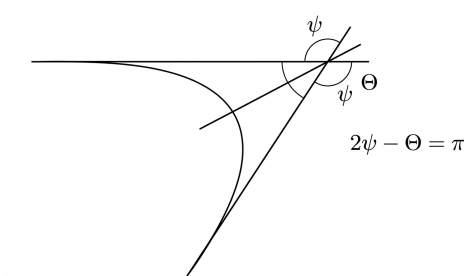
that case, we could have deflection through the same angle for different values of b . Then our expression for $\sigma(\Theta)$ becomes

$$\sigma(\Theta) = \sum_{\alpha} \frac{b_{\alpha}}{\sin \Theta} \left| \frac{db}{d\Theta} \right|_{\alpha}$$

where α is sum over each monotonic interval of b_{α} . Note that at $\Theta \rightarrow \Theta_m$, $\sigma(\Theta)$ diverges, since $\left. \frac{d\Theta}{db} \right|_{\Theta_m} = 0$. This is known as rainbow scattering.

Attractive Potentials

For attractive potentials the incoming particle will get deflected toward the centre of the potential.

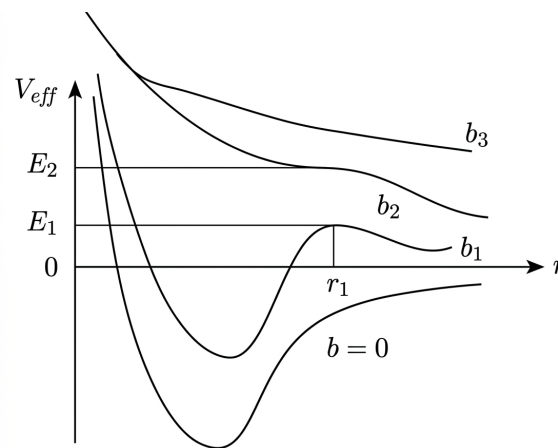


For such cases the angle Ψ may be greater than $\frac{\pi}{2}$ and so Θ would be defined by the absolute value $\Theta = |\pi - 2\Psi|$. However for suitable potentials the angle Θ can be greater than 2π which reflects the situation in which a particle can 'orbit' the centre once or more before escaping.

To see how this may happen we consider the effective (angular momentum-dependent) potential, known as the scattering potential. Let us consider a case where the original potential is attractive but falls off faster than $\frac{1}{r^2}$. Since the l -dependent term in V_{eff} is of the form:

$$\frac{l^2}{2mr^2} = \frac{Eb^2}{r^2} \quad [\text{using } l = b\sqrt{2mE}]$$

If the attractive potential falls off faster than $\frac{1}{r^2}$ then one can have effective potentials of various forms with local maxima/minima.



In such cases, let us consider the potential for which $b = b_1$ and we have a 'bump' at $r = r_1$. For E exactly equal to $V_{\text{eff}}(r_1)$ we will have a circular orbit. But such an orbit is unstable and highly unlikely. But for E roundabout that value $\dot{r}$ becomes very small and the particle spends a long time near $r = r_1$. During this time $\dot{\theta}$ is given by

$$\dot{\theta} = \frac{l_1}{mr_1^2} = \frac{b_1}{r_1^2} \sqrt{\frac{2E}{m}}$$

may be significant and the particle may do one or more revolutions before escaping again. This phenomenon is known as orbiting.

For E greater than $V_{\text{eff}}(r_1)$ the scattering will be dominated by the repulsive part of the potential.

For the case when $b = b_2$ there is an inflection point and for energy matching that point the particle may experience little or no deflection. This is due to the fact that at the inflection point $b = b_2$ of the impact factor the potential is both repulsive and attractive at the same time.